100 Kubernetes Errors

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What Breaks, Why It Breaks, and How to Fix it

	Kubernetes Error	Cause	Fix
01	❌ **CrashLoopBackOff**	App crash or incorrect command	Check logs, fix entrypoint or config
02	❌ **ImagePullBackOff**	Wrong image name or secret issue	Verify image name, registry creds
03	❌ **ErrImagePull**	Image doesn't exist or inaccessible	Correct the image URL or credentials
04	❌ **CreateContainerConfigError**	Invalid env or config	Check ConfigMap/Secret references
05	❌ **CreateContainerError**	Issue during container creation	Check image health, volume mounts
06	❌ **RunContainerError**	App failed to run post-creation	Inspect logs, entrypoints, permissions
07	❌ **OOMKilled**	Container exceeded memory limit	Tune limits, optimize memory usage
08	❌ **Back-off restarting failed container**	Container repeatedly fails	Check logs, increase backoff, fix bug
09	❌ **NodeNotReady**	Node down or unreachable	Restart node, check kubelet, infra issues
10	❌ **PodUnschedulable**	Insufficient resources or constraints	Adjust node pools, affinities, quotas

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11	❌ **NodeAffinity conflict**	Pod's affinity can't be met	Reconfigure node labels or affinities
12	❌ **PVC Pending**	No matching PV available	Provision PV manually or enable dynamic provisioning
13	❌ **VolumeMountError**	Mount path or volume issue	Check access modes, volume specs
14	❌ **DeadlineExceeded**	Job exceeded deadline	Increase deadline or optimize logic
15	❌ **InvalidImageName**	Malformed image reference	Use valid, versioned image names
16	❌ **Secret not found**	Missing secret reference	Create the required secret
17	❌ **ConfigMap not found**	Missing config map reference	Create or correct reference
18	❌ **NodeSelector mismatch**	No matching node for selector	Adjust labels/selectors
19	❌ **Invalid container port**	Declared port not exposed	Match container and service port
20	❌ **FailedScheduling**	Pod can't find suitable node	Check taints, resources, affinity
21	❌ **Liveness probe failed**	Health check endpoint fails	Fix probe or app logic
22	❌ **Readiness probe failed**	Pod not ready to serve	Fix app or probe definition
23	❌ **Service not reachable**	Incorrect selector or port	Fix service spec or target labels
24	❌ **DNS resolution failed**	CoreDNS crash or config issue	Restart CoreDNS, fix DNS config
25	❌ **LoadBalancer pending**	Cloud LB not provisioned	Check cloud integration and permissions

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26	❌ **RBAC access denied**	Insufficient user permission	Add proper roles via RoleBinding
27	❌ **Crash in InitContainer**	Init script failed	Fix init container logic or script
28	❌ **ImagePullSecrets error**	Invalid registry secret	Fix or re-add secret
29	❌ **Invalid PVC access mode**	Mismatch with PV mode	Match ReadWriteOnce vs Many
30	❌ **Unbound PersistentVolumeClaim**	PVC can't bind to PV	Provision matching PV
31	❌ **Pod evicted**	Node pressure (memory, disk)	Scale nodes, tune resource limits
32	❌ **Unauthorized error**	Token or cert expired	Renew credentials
33	❌ **Admission webhook denied pod**	Policy validation failed	Fix policy or admission config
34	❌ **PodSecurityPolicy violation**	Spec disallowed by PSP	Adjust policy or pod spec
35	❌ **Pod quota exceeded**	Namespace resource quota hit	Raise limits or clean up pods
36	❌ **API throttling**	Too many API requests	Use client-side throttling or retries
37	❌ **Invalid label format**	Label uses invalid characters	Fix label to match regex
38	❌ **Service endpoints not populated**	No matching pods	Check pod selectors
39	❌ **Helm release failed**	Chart or values error	Validate templates, use dry-run
40	❌ **Helm rollback failed**	Previous state corrupt	Manually delete or fix resources

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41	❌ **Node disk full**	Pod logs or volumes overfill node	Use log rotation, cleanup jobs
42	❌ **Invalid YAML**	Syntax error in manifest	Lint and validate YAML
43	❌ **Container exits immediately**	No running command	Use proper entrypoint or CMD
44	❌ **Job doesn't complete**	Hangs or fails silently	Set backoffLimit, add alerting
45	❌ **DaemonSet pods not scheduled**	Taints not tolerated	Add tolerations to spec
46	❌ **Pod can't mount ConfigMap**	Volume not defined correctly	Fix volumeMounts and volumes
47	❌ **Invalid node port range**	Outside cluster port range	Use ports within 30000â€“32767
48	❌ **Pod terminated with exit code 137**	OOM or SIGKILL	Check logs, increase memory
49	❌ **Job history limits too high**	Jobs clog API server	Set history limits
50	❌ **Pod never terminates**	Container runs forever	Add timeout or readiness check
51	❌ **Invalid annotation key**	Violates key naming rules	Fix annotation format
52	❌ **Deployment rollout stuck**	Unavailable replicas > threshold	Check pod health and quotas
53	❌ **Service without selector**	No endpoints generated	Manually define endpoints or add selector
54	❌ **Container can't write logs**	No log path or permissions	Set proper path or mount volume
55	❌ **HorizontalPodAutoscaler not scaling**	Missing metrics-server	Install metrics-server

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56	❌ **Deployment loops without change**	Incorrect label match or template hash	Use proper selectors
57	❌ **Pod IP not reachable**	CNI plugin issue	Check CNI logs and restart plugin
58	❌ **ClusterIP service unreachable**	IPTables issue	Restart kube-proxy or check rules
59	❌ **Controller-manager crashloop**	Certs or leader election config	Check config and logs
60	❌ **Too many open files**	File descriptor limit hit	Increase ulimit on container
61	❌ **Pod fails DNS resolution inside cluster**	kube-dns not resolving svc.cluster.local	Restart CoreDNS and validate config
62	❌ **HPA flapping up/down**	Thresholds too tight	Add cool-down and buffer ranges
63	❌ **Node cordoned unexpectedly**	Drift or manual action	Uncordon or automate via scripts
64	❌ **Evicted due to node reboot**	No pod disruption budget	Add PDB to critical workloads
65	❌ **Service has no endpoints**	Pods not matching selectors	Fix labels or selectors
66	❌ **Service has wrong targetPort**	Service port doesn't match container	Correct service spec
67	❌ **Pod fails due to image tag `latest`**	Old image cached	Use versioned tags
68	❌ **Kubelet not registering node**	Firewall or DNS block	Open ports, validate DNS
69	❌ **API server not reachable**	Network or cert issue	Check control plane and network
70	❌ **Pod takes too long to start**	Init delay, heavy image	Optimize image size and probes

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71	❌ **Pod fails due to PodToleratesNodeTaints**	Taints on the node not tolerated by pod	Add appropriate tolerations to pod spec
72	❌ **failed to pull image but image exists**	Image pull secrets misconfigured	Verify and fix imagePullSecrets
73	❌ **Ingress returns 404 despite healthy service**	Wrong path or missing backend rule	Check Ingress path config and backend mapping
74	❌ **Persistent Volume stuck in Terminating**	Finalizer blocks deletion	Manually remove finalizer or force delete
75	❌ **Readiness probe fails but container healthy**	Probe path or port incorrect	Correct the readiness probe URL/port
76	❌ **Pods stuck due to affinity rules**	Overly strict affinity/anti-affinity	Loosen affinity rules or add tolerations
77	❌ **Node pressure eviction**	DiskPressure or MemoryPressure status	Clean up disk, scale node pool, or adjust limits
78	❌ **No routes found in Calico networking**	BGP misconfigured or broken	Restart Calico nodes or reset BGP config
79	❌ **rpc error: code = Unknown on API calls**	gRPC call from controller failed unexpectedly	Restart API-dependent controllers
80	❌ **Failed attaching volume: Multi-Attach error**	Volume used by another pod	Ensure volume supports ReadWriteMany or detach
81	❌ **sysctl not allowed**	Pod uses unsupported sysctls	Add allowed sysctls to kubelet config
82	❌ **hostname unknown errors in containers**	Missing DNS resolution or CoreDNS crash	Restart CoreDNS, validate resolv.conf
83	❌ **CrashLoopBackOff due to failed initContainer**	Init container failed, main never starts	Fix or remove the initContainer logic
84	❌ **volumeMount fails due to read-only mode**	Mount declared readOnly but write attempted	Set readOnly: false if writes are needed
85	❌ **NetworkPolicy blocks egress to internet**	Deny-all policy in place	Allow required CIDRs or specific FQDNs

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86	❌ **Secret not found during deploy**	Namespace mismatch or race condition	Ensure secret exists in correct namespace
87	❌ **ConfigMap is immutable error**	Trying to update immutable ConfigMap	Delete and recreate the ConfigMap
88	❌ **Pod stuck Pending due to NoVolumeZoneConflict**	PVC requested a zone not available on node	Use zonal PVCs or schedule to matching zones
89	❌ **API server connection refused**	Broken kubeconfig or API not running	Verify kubeconfig context and cluster status
90	❌ **Sidecar container causes pod to never terminate**	Main container exits, sidecar runs forever	Use sidecar lifecycle hooks to terminate
91	❌ **HPA not scaling pods**	Metrics server not reporting	Install/verify metrics-server
92	❌ **kubectl get hangs indefinitely**	API server overloaded or DNS misconfig	Restart CoreDNS or check master node health
93	❌ **Logs not available after pod crash**	Ephemeral storage full or log rotation	Use centralized logging or persistent logs
94	❌ **cert expired on Kubelet or API server**	Cluster certificate not rotated	Renew via kubeadm certs renew or automation
95	❌ **Webhook validation timeout**	Webhook backend slow or misconfigured	Fix webhook service or increase timeout
96	❌ **PodDisruptionBudget blocks rollout**	Too few pods allowed to disrupt	Adjust PDB thresholds during rollout
97	❌ **Pod stuck due to unresolved hostPath mount**	Host path missing on node	Ensure host path exists or use PV instead
98	❌ **containerCreating forever with sandbox issue**	Containerd or Docker sandbox stuck	Restart container runtime or node
99	❌ **Kube-proxy fails on startup**	IPTables rules corrupted	Flush and restore iptables or reboot node
100	❌ **QoS mismatch causes unexpected eviction**	Wrong limits/requests combo	Use guaranteed/burstable classes intentionally